

PAC.

Given Data.

- Vexpected (Total Expected Vehicles)
- Cap A = 20
Cap B = 40
Cap C = 15
- for each vehicles
 1. VType (vehicle type)
 2. Vcat (User Category)
 3. Permit?
 4. emergency.

Required Results

- Real time Vehicle Accepted and Rejected Messages
- Remaining Zone Capacities
- Summary AT the end of Parking
 1. Total Processed
 2. Total Accepted
 3. Total Rejected.
 4. Total number of Cars, Bikes & Vans Parked
 5. final occupancy & Remaining spaces of each zone
 6. Zone with Highest occupancy
 7. Campus parking full or Not.

Processing Required.

- Use for loop for 1 to $V_{expected}$
- Validate All inputs.
- Determine target Zone and Spaces needed.
 1. 2 for Vans
 2. 1 for others
- check available spaces by. $CapX - OccX \geq Spaces$
- Update Occupancy at Zones and Altered and Rejected Respectively
- Identify Zone with Highest Occupancy.
- Verify if Campus Parking is full or not.

Alternative Solutions

- Reject immediately. if $Permit = N$ & $emergency = N$.

- Reject if target Zone is full

check Highest Occupancy by:
~~Zone A & Zone B, Zone A~~
 Zone C

- check if 2 zones Occupancy are full or All 3 are equal.